

CAIR2 Integration Project

Complete Technical Guide — ClaudMD / Aeliusmd

HL7 v2.5.1 VXU | California Immunization Registry | Version 1.0

1. Project Overview

This project sends vaccination records from ClaudMD clinic EMR databases to **CAIR2** (California Immunization Registry). CAIR2 is the statewide system that stores immunization history for California residents. Healthcare providers are required to report vaccinations electronically using standard HL7 messages.

What we build: A Python background service that reads vaccination data from each clinic database, converts it to an HL7 VXU message, submits it to CAIR2 via SOAP, and records success or failure for retries.

FLOW:

```
graph TD
    A[Nurse saves vaccination in ClaudMD (EHRVaccines table)] --> B[Queue record created (outbox table OR IsSubmitted = 0)]
    B --> C[cair_service.py runs every 5 minutes]
    C --> D[Coordinator reads master DB (ClinicSetup) for active clinics]
    D --> E[Worker pool (3 threads) processes batches per clinic]
    E --> F[Build HL7 VXU message from patient + vaccine data]
    F --> G[POST to CAIR SOAP endpoint (submitSingleMessage)]
    G --> H[Parse ACK response: AA=accepted, AE=retry, AR=rejected]
```

2. Your CAIR Onboarding Details

Setting	Value
Organization	Aeliusmd Inc
CAIR Org Code	SF-013259
Contact email	CAIRDataExchange@cdph.ca.gov
Spec document	CAIR2_HL7v2.5.1DataExchangeSpecs.pdf v3.10
Message type	VXU (Unsolicited Vaccination Update)
HL7 version	2.5.1 only
Transport	SOAP web service (not MLLP/TCP)

SOAP Endpoints

Environment	URL / Operation
ONBOARDING (test first)	Endpoint: https://cdph-interop-stage.cdph.ca.gov/services/client_Service.client_ServiceHttpSo
WSDL (stage)	https://cdph-interop-stage.cdph.ca.gov/CASTG-WS/IISService?WSDL
PRODUCTION	Endpoint: https://cdph-interop-prod.cdph.ca.gov/services/client_Service.client_ServiceHttpSo
WSDL (prod)	https://cdph-interop-prod.cdph.ca.gov/CAPRD-WS/IISService?WSDL
VXU operation	submitSingleMessage
Query operation	submitSingleQuery (QBP/RSP - optional)
Old endpoint (DO NOT USE)	https://cair.cdph.ca.gov/CAPRD-WS/IISService

3. HL7 VXU Message Structure

A VXU message is a pipe-delimited (|) text document. Each line is a segment. Every vaccination submission must be a **full VXU** with all required fields, even when updating an existing record.

Segment	Required	Purpose
MSH	Yes	Message header: sender, receiver, timestamp, message type
PID	Yes	Patient: name, DOB, sex, address, phone/email
PD1	Yes	Privacy: protection indicator, publicity code
NK1	Optional	Next of kin / guardian
ORC	Yes	Order control: links vaccination to check-in
RXA	Yes	Vaccine administration: code, dose, lot, date
RXR	Optional	Route and body site (e.g. IM, left arm)
OBX	Optional	Observations: VFC funding eligibility

Official CAIR Sample Message (from cair emails.txt)

```
MSH|^~\&||CA0012345||CAIR2|20260816023022-0700||VXU^V04^VXU_V04|
3f2504e0-4f89-11d3-9a0c-0305e82c3301|P|2.5.1|||AL|AL|||Z22^CDCPHINVS|CA0054321
PID|1||AC10293^^^CA0012345^MR||Alvarez^Maria^^^^^|^^^^^^|19880412|F||
2106-3^^CDCREC|123 Main St^^Sacramento^CA^95814||
^NET^Internet^maria.alvarez@example.com~^PRN^CP^^^916^5550134||^|
2186-5^^CDCREC||
PD1|||||||^^N|20260810||||
ORC|RE|48213^CMC|48213^CMC||||
RXA|0|1|20260810||00069-1000-03^NDC|0.3|mL^^UCUM||
00^NEW IMMUNIZATION RECORD^NIP001||^CA0054321||FA7205||PFR^^MVX||CP|A|
20260816023022-0700
OBX|1|CE|64994-7^Vaccine funding program eligibility category^LN|1|
V01^Not VFC eligible^HL70064||||F||20260810
```

4. Complete HL7 Field Reference

MSH - Message Header

Field	Usage	Value / Example	Notes
MSH-1	R		Field separator
MSH-2	R	^~\&	Encoding characters
MSH-4	R	SF-013259	Sending facility = your CAIR org code
MSH-6	R	CAIR2	Receiving facility
MSH-7	R	20260816023022-0700	Message datetime

Field	Usage	Value / Example	Notes
MSH-9	R	VXU^V04^VXU_V04	Message type
MSH-10	R	GUID	Unique message control ID
MSH-11	R	P or T	P=production, T=training/test
MSH-12	R	2.5.1	HL7 version - only 2.5.1 accepted
MSH-15	RE	AL	Accept acknowledgment type
MSH-16	RE	AL	Application acknowledgment type
MSH-22	RE	SF-013259	Responsible org / site where vaccine given

PID - Patient Identification

Field	Usage	ClaudMD Source	Example
PID-3	R	Patients.AccountNumber	AC10293^^^SF-013259^MR
PID-5	R	Patients.LastName, FirstName	Alvarez^Maria
PID-7	R	Patients.DateOfBirth	19880412
PID-8	R	Patients.GenderId (lookup)	F / M / X / U
PID-11	RE	Address1, City, State, ZipCode	123 Main St^^Sacramento^CA^95814
PID-13	RE	CellPhone, HomePhone, Email	See PID-13 section below
PID-22	RE	EthnicityId (lookup)	2186-5^^CDCREC

RXA - Vaccine Administration

Field	Usage	ClaudMD Source	Example
RXA-1	R	Hardcoded	0
RXA-2	R	Hardcoded	1
RXA-3	R	EHRVaccines.VaccineDate	20260810
RXA-5	R	ServiceCodes.NDCNumber or CVX	X00069-1000-03^^NDC
RXA-6	R	EHRVaccines.Dosage	0.3 (use 999 if unknown)
RXA-7	R	Hardcoded if dose known	mL^^UCUM
RXA-9.1	R	Hardcoded	00 = given shot (not historical)
RXA-11.4	R	CAIR site ID	SF-013259

Field	Usage	ClaudMD Source	Example
RXA-15	R	EHRVaccines.LotNumber	FA7205
RXA-16	RE	EHRVaccines.VaccineExpirationDate	20270531
RXA-17	R	EHRVaccines.Manufacturer / MVX	PFR^^MVX
RXA-20	RE	Hardcoded	CP = complete
RXA-21	RE	Hardcoded	A = add new record
OBX-5	R	VFC eligibility code	V01^Not VFC eligible^HL70064

5. PID-13 Contact Information (CRITICAL)

CAIR requires at least one of: home phone, cell phone, or email. Messages without valid contact info receive warnings or rejection:

```
ERR||PID^1^13|101|Warning: Home Phone or Cell Phone is missing or invalid
ERR||PID^1^13|102|Warning: Invalid Client Email
```

Contact Type	PID-13.2	PID-13.3	PID-13.4	PID-13.6	PID-13.7
Home phone	PRN	PH	-	area code	number
Cell phone	PRN	CP	-	area code	number
Email	NET	Internet	email@address.com	-	-

PID-13 Format Examples

Cell phone only:

```
^PRN^CP^^^916^5550134
```

Email + cell phone:

```
^NET^Internet^maria.alvarez@example.com~^PRN^CP^^^916^5550134
```

Home + email + cell:

```
^PRN^PH^^^213^5555555~^NET^Internet^email@x.com~^PRN^CP^^^213^5551234
```

QA database finding: Patients table has CellPhone populated but Email is mostly NULL. Update patient intake to collect email or ensure cell phone is always present.

6. Vaccine Codes Explained

Code	Full Name	Used In	Example	Purpose
CVX	CDC Vaccine Code	RXA-5	08=Hep B, 207=COVID	Clinical vaccine identity
MVX	Manufacturer Code	RXA-17	PFR=Pfizer, MSD=Merck	Who made the vaccine
NDC	National Drug Code	RXA-5	00069-1000-03	Product package ID
CPT	Billing Code	NOT in VXU	90471	Reimbursement only
HL70064	VFC Eligibility	OBX-5	V01, V03	Funding category

Important: CAIR accepts CVX **or** NDC in RXA-5, not both together. CPT codes are for billing and are NOT used as the primary vaccine identifier in CAIR.

Fields That Must Be Added to ServiceCodes (from CAIR email)

Field	Type	Priority	Description
CvxCode	varchar(10)	MUST HAVE	CDC vaccine code. String not int (e.g. 09)

Field	Type	Priority	Description
MvxCode	varchar(10)	MUST HAVE	Manufacturer code e.g. SKB, PFR
IsVaccineProduct	bit	MUST HAVE	Exclude admin fees from reporting
RegistryApprovedOn	date	MUST HAVE	Null = do not report to CAIR
RouteCode	varchar(20)	SHOULD HAVE	EC28161=IM. CAIR rejects without route
DoseAmount	decimal(9,3)	SHOULD HAVE	E.g. 0.5
DoseUnit	varchar(12)	SHOULD HAVE	E.g. mL
Ndc11	varchar(11)	SHOULD HAVE	11-digit NDC (10-digit is ambiguous)
CvxText / MvxText	varchar(200)	SHOULD HAVE	Human-readable labels

7. ClaudMD Database Structure

Master Database: ClaudMD_QA_Setup

Server: 10.103.0.201 | Database: ClaudMD_QA_Setup

This is the **clinic registry**, not the EMR data. It lists all clinics and their individual database connection details.

ClinicSetup Column	Purpose
ClinicID	Internal clinic identifier
ClinicName	Display name
DatabaseServer	SQL Server for this clinic
DatabaseName	Clinic EMR database name
DatabaseUser / DatabasePassword	Connection credentials
ActivationKey	e.g. 20000002
Active	1 = clinic is active

Active QA Clinics (from database read)

ActivationKey	ClinicName	DatabaseName
20000001	ClaudMD_VCOMC_QA_1	ClaudMD_VCOMC_QA_1
20000002	ClaudMD_VCOMC_QA_2	ClaudMD_VCOMC_QA_2
20000003	ClaudMD_VCOMC_QA_3	ClaudMD_VCOMC_QA_3
20000004	ClaudMD_Medcube_QA	ClaudMD_Medcube_QA
20000005	Aelius_vcomc_QA	Aelius_vcomc_QA

Clinic EMR Database Tables (421 tables per clinic)

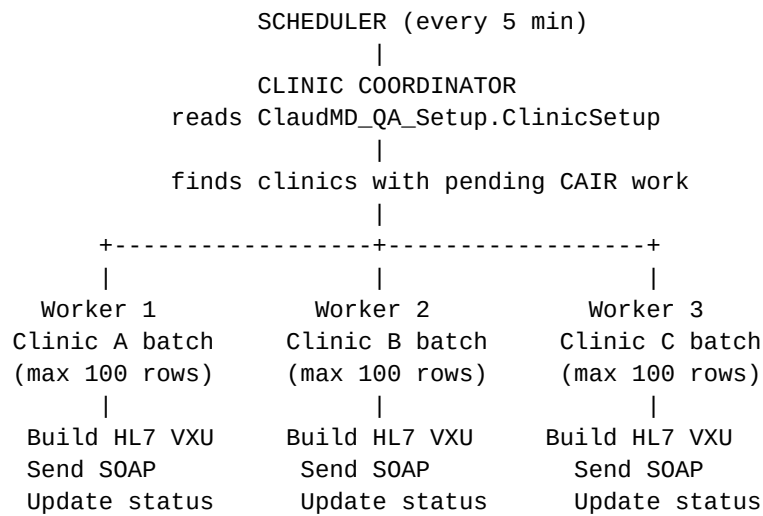
CAIR Concept	ClaudMD Table	Key Columns
Patient	dbo.Patients	AccountNumber, FirstName, LastName, DateOfBirth, GenderId, CellPh
Check-in / Visit	dbo.CheckInsHeader	Id, PatientId, CheckInDate, CheckInTime, ProviderId
Vaccination	dbo.EHRVaccines	CheckInId, ServiceCodeId, VaccineDate, LotNumber, Dosage, Route, I
Service / Product	dbo.ServiceCodes	Code, Description, CPTCode, NDCNumber
Billing charges	dbo.Transactions	CheckInId, ServiceCodeId, Units, NDCNumber

EHRVaccines - Full Column List

Id, CheckInId, ServiceCodeId, OrderServicesId, VaccineName, LotNumber, VaccineExpirationDate, Dosage, VaccinatorId, BodySite, Route, Manufacturer, VaccineDate, VaccineTime, IsVaccine, IsSubmitted, IsDeleted, Dose, VisPublicationDate, VaccineNotDoneReasonId, Remarks

Key flags: IsVaccine=1 means real vaccine (not TB test etc). IsSubmitted=0 means not yet sent to CAIR. QA currently has 24 EHRVaccines rows but 0 with IsVaccine=1 and 0 with lot numbers filled.

8. Worker Architecture (Multi-Clinic)



Outbox Status Flow

Status	Meaning	Next Action
PENDING	New vaccination queued	Worker picks up on next run
PROCESSING	Worker claimed record	Building/sending HL7
SENT	CAIR accepted (ACK=AA)	Done
RETRY	Temporary failure (timeout/network)	Retry after backoff delay
FAILED	Permanent error (bad data)	Manual review required

Retry Schedule (Exponential Backoff)

Attempt	Wait Before Retry
1	5 minutes
2	15 minutes
3	30 minutes
4	60 minutes
5	120 minutes
6+	Mark as FAILED

Multi-clinic rules: Workers are a shared pool (not 1 worker per clinic forever). Only one active batch per clinic at a time to avoid race conditions. Retries and new PENDING records share the same queue.

9. Project Files & Setup

File	Purpose
cair_service.py	Main background worker - run continuously
demo_build_vxu.py	Generate sample HL7 without database
scripts/seed_sithum_cair_test_data.py	Dev: insert test vaccines + CAIR queue rows
run_dry_once.py	One worker cycle — read DB, build HL7 (dry-run)
cair_integration/hl7/vxu_builder.py	Builds HL7 VXU from data models
cair_integration/cair/soap_client.py	Sends HL7 to CAIR SOAP endpoint
cair_integration/cair/ack_parser.py	Parses ACK response (AA/AE/AR)
cair_integration/worker/coordinator.py	Multi-clinic job dispatcher
cair_integration/worker/processor.py	Processes one outbox record
cair_integration/outbox/repository.py	Database access layer
cair_integration/vaccination_service.py	Creates outbox on vaccine save
.env	Credentials and config (never commit to git)
sql/schema.sql	Outbox table DDL (optional - adapt for ClaudMD)

Environment Variables (.env)

QA mode — MASTER_DB_* + ClinicSetup. **Development mode** — CLINIC_DB_* direct to Sithum clinic DB (skips ClinicSetup).

```
# Development (Sithum clinic DB)
APP_ENV=development
CLINIC_DB_SERVER=10.103.0.211
CLINIC_DB_NAME=ClaudMD_Development_Sithum
CLINIC_DB_USER=testuser
CLINIC_DB_PASSWORD=***

# QA (master DB + ClinicSetup)
MASTER_DB_SERVER=10.103.0.201
MASTER_DB_NAME=ClaudMD_QA_Setup
MASTER_DB_USER=testuser
MASTER_DB_PASSWORD=***
DEFAULT_ACTIVATION_KEY=20000002

CAIR_SOAP_URL=https://cdph-interop-stage.cdph.ca.gov/services/
  client_Service.client_ServiceHttpSoap12Endpoint
CAIR_SOAP_USERNAME=  # required to send HL7
CAIR_SOAP_PASSWORD=  # required to send HL7
SENDING_FACILITY_ID=SF-013259
RECEIVING_FACILITY=CAIR2
WORKER_BATCH_SIZE=100
WORKER_POOL_SIZE=3
SCHEDULER_INTERVAL_SECONDS=300
```

Commands to Run

```
pip install -r requirements.txt
python scripts/seed_sithum_cair_test_data.py seed      # dev test data
python scripts/seed_sithum_cair_test_data.py verify   # check queue
python demo_build_vxu.py                             # Test HL7 without database
python run_dry_once.py                               # Build HL7 from DB (no CAIR send)
python cair_service.py                               # Start background worker
```

Development Testing (Sithum DB)

The seed script inserts EHRVaccines and EHRVaccineThirdPartySubmissions for testing. **It does not send HL7.** SubmitStatus stays 0 and RequestPayload is empty until cair_service.py runs with valid CAIR SOAP credentials.

Seed command	Purpose
list	Show published visits and existing submissions
seed	Create test vaccines + queue rows (SubmitStatus=0)
verify	Confirm worker can pick up pending rows
reset	Reset LOT-CAIR-TEST-* rows back to PENDING

10. ACK Response Handling

After submitting a VXU, CAIR returns an HL7 ACK (acknowledgment) message:

```
MSH|^~\&|CAIR2|CAIRLO|MyEMR|SF-013259|...|ACK^V04^ACK|{message_id}|P|2.5.1  
MSA|AA|{message_id}  
ERR|... (only if errors)
```

MSA-1 Code	Meaning	Action
AA	Application Accept	Mark outbox as SENT
AE	Application Error	Schedule RETRY (temporary)
AR	Application Reject	Mark as FAILED (fix data)

11. Inventory Decrementing (Optional CAIR Feature)

CAIR can automatically decrement vaccine inventory when a shot is reported. This is OFF by default. Contact CAIR Data Exchange team to enable.

Field	Requirement
MSH-22	Must match CAIR site ID where inventory is stored
RXA-5.1	Vaccine code must match inventory
RXA-9.1	Must be 00 (given shot, not historical)
RXA-11.4	Must match MSH-22 site ID
RXA-15	Lot number must match CAIR inventory
RXA-20	CP, PA, or empty
RXA-21	A or U
OBX-5.1	VFC funding must match lot funding category

12. Current Blockers & Action Items

#	Blocker	Owner	Action
1	No CvxCODE/MvxCODE in ServiceCenter table	EMR team	Add columns + CDC lookup tables
2	Patient email mostly NULL	EMR team	Require email or cell at intake
3	Lot number never filled	Clinical staff	Make mandatory on vaccine entry
4	Route field empty	EMR team	Add RouteCode default per vaccine product
5	No test vaccine data (IsVaccine=0)	QA team	Create test vaccination records

#	Blocker	Owner	Action
6	NDC empty on vaccine codes	EMR team	Populate NDC or use CVX mapping
7	Not tested on CAIR stage	Dev team	Submit test VXU, email CAIR team
8	GenderId is int not M/F	Dev team	Add lookup GenderId to HL7 sex code

13. Testing Checklist

- Configure .env — CLINIC_DB_* for Sithum dev or MASTER_DB_* for QA
- Run scripts/seed_sithum_cair_test_data.py seed then verify (dev DB)
- Confirm SubmitStatus=0 and RequestPayload empty — no HL7 sent yet
- Configure .env with stage SOAP endpoint, credentials, and SF-013259
- Run demo_build_vxu.py and verify HL7 output matches CAIR sample format
- Run run_dry_once.py to build HL7 from real DB rows without sending
- Create test patient with CellPhone AND Email populated
- Create EHRVaccines record with IsVaccine=1, LotNumber, Route, Dosage filled
- Add CVX/MVX codes to the linked ServiceCode record
- Submit test message to CAIR stage endpoint (cair_service.py)
- Email CAIRDataExchange@cdph.ca.gov to request review of test submissions
- Fix any ERR warnings in ACK response
- Switch to production endpoint after CAIR approval
- Deploy cair_service.py on server with scheduler running every 5 minutes

Document generated from: CAIR2_HL7v2.5.1DataExchangeSpecs.pdf, Required Fields mapping Excel, cair emails.txt, ClaudMD QA database inspection, and cair_integration Python implementation.